Dynamics of Mitochondrial Transfer in Tumors: Computational Pipeline using MERCI & Cell Ranger

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Abstract

Mitochondrial transfer within the tumor microenvironment (TME) is a critical mechanism influencing cancer progression and immune evasion. Cancer cells may acquire healthy mitochondria from infiltrating immune cells to restore oxidative phosphorylation (OXPHOS), while in other cases, tumor-derived mitochondria are transferred to immune cells, impairing their function.
We reproduced the MERCI pipeline to computationally infer mitochondrial transfer using mixed mitochondrial variant signatures. We additionally constructed an end-to-end workflow to convert raw SRA files into aligned BAM files using FASTQ extraction and Cell Ranger. We run the MERCI algorithm on co-cultured samples of macrophages and MDA-MB-231 cells. We find heterogenous patterns of mitochondrial transfer in both samples.

Introduction

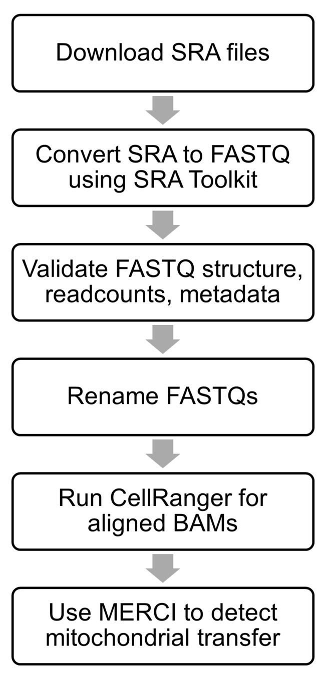
Mitochondria regulate apoptosis, ROS balance, metabolic adaptation, and immune signaling. In solid tumors, mitochondria frequently move between cancer cells and T cells via tunneling nanotubes, cell fusion, or EVs.

Two major transfer modes have been reported:

- Cancer cells acquire healthy mitochondria from T cells (restoring OXPHOS)
- Cancer cells export mutated mitochondria to T cells (reducing immune function)

Understanding transfer directionality is essential for interpreting tumor immune escape and therapeutic response.

Overall Computational Pipeline



Stages:

1. Download SRA files
2. Convert SRA → FASTQ using SRA Toolkit
3. Validate FASTQ structure, read counts, metadata
4. Rename FASTQs to Cell Ranger format
5. Run Cell Ranger to generate aligned BAMs
6. Use MERCI to detect mitochondrial transfer

SRA to FASTQ Pipeline

Input: SRA IDs (SRR15339529,SRR15339531) **Output:** Paired FASTQs (_1.fastq.gz, _2.fastq.gz)

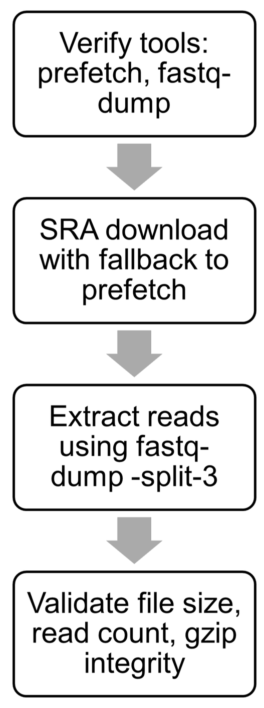
FASTQ structure:

$$N_{\text{reads}} = \frac{L}{4}$$

where L = total number of lines in the FASTQ file.

Major Steps:

- Verify tools: `prefetch`, `fastq-dump`
- Direct AWS SRA download with fallback to prefetch
- Extract reads using `fastq-dump -split-3`
- Validate file size, read count, gzip integrity



FASTQ to BAM Using Cell Ranger

Input: Correctly renamed FASTQs as per bcl2fastq format, hg38 human reference genome **Output:** BAM file

Cell Ranger steps:

1. STAR read alignment:

$$\hat{x} = \arg\max_{x \in G} S(read, x)$$

2. Barcode correction via Hamming distance:

$$d_H(b_{\text{read}}, b_{\text{valid}}) \leq 1$$

3. UMI collapsing:

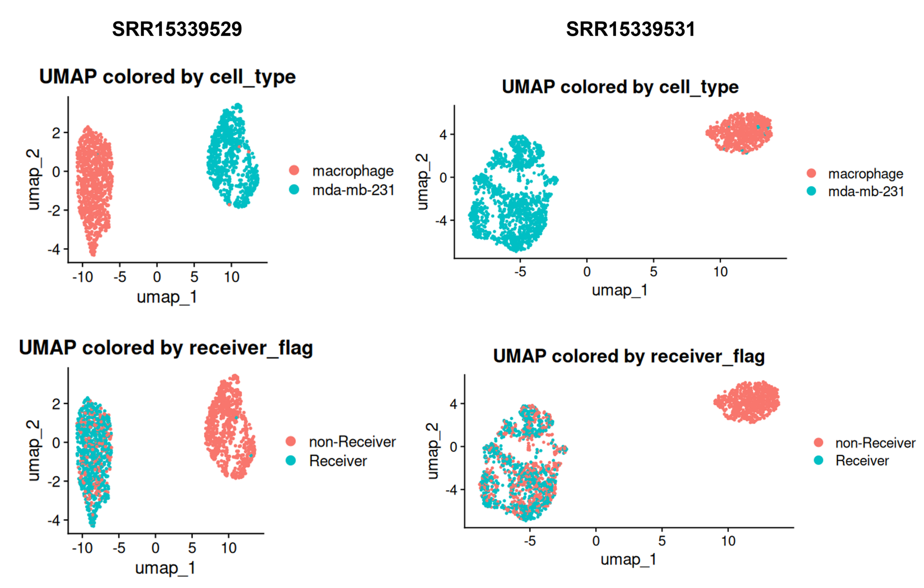
$$\text{UMI}_{unique} = \text{unique}(barcode, UMI, gene)$$

References

Zhang et al., MERCI Framework (2023)
Strating et al., Co-cultures of colon cancer (2023)
Hong et al., Tumor Microenvironment (2023)
Del Rio et al., Mitochondrial Transfer (2024)
Berridge et al., Horizontal mitochondrial transfer (2025)

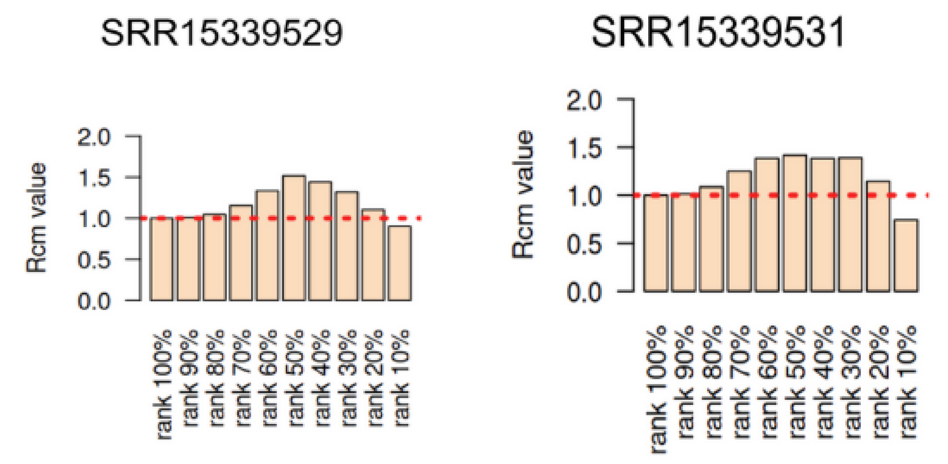
Results

- SRR15339529 shows highly specific mitochondrial transfer into macrophages
- SRR15339531 shows a more diffuse transfer signal across both macrophages and MDA-MB-231 cells.



Relative copy number of mitochondria (Rcm) distribution across cells in each sample.

- SRR15339529 shows elevated Rcm in a focused subset of cells, consistent with targeted mitochondrial transfer into macrophages.
- In contrast, SRR15339531 displays a broader elevation pattern, supporting a more diffuse transfer signal across both macrophages and MDA-MB-231 cells.



Discussion

We find that mitochondrial transfer is highly cell-type specific in SRR15339529, with macrophages acting as the primary receivers, while SRR15339531 shows broader uptake across both macrophages and MDA-MB-231 cells. This demonstrates that MERCI can sensitively capture heterogeneous transfer behaviors across samples.

Future Directions:

- Given our observation of sample-dependent heterogeneity, future studies should examine mitochondrial transfer in more microenvironmental settings
- MERCI can be integrated with clinical data to explore associations with tumor grade, cancer progression, and patient prognosis.
- Apply MERCI to other tumor types